1	2	3	4	5	6	Total
/20	/20	/20	/20	/20	/20	/100

Name: ________________________________ May 11, 2013 Saturday

Department/School:______________________ SID:____________________________

University Physics

Midterm Examination

Kuang Yaming Honors School, Nanjing University

Select five out of following six problems.

1. (20pts) A particle moves in a one-dimensional space. The potential energy of the particle is given by

$$U(x) = 6x^2 - 2x^3.$$

(a) Determine the force acting on the particle.
(b) At what positions is the particle in equilibrium?
(c) Which of these equilibrium positions are stable and which are unstable?

2. (20pts) A ball of mass m moves under the uniform gravity and in a medium that exerts a resistance force $f_r = Cv^4$ on it. When the ball falls vertically, its terminal speed is v_0.

(a) Express the coefficient C in terms of gravity g, mass m and terminal speed v_0.
(b) Write the equation of motion for the upward motion of the ball.
(c) If the ball is projected vertically upward with an initial speed u, find the maximum height the ball can reach in terms of g, v_0 and u.
(d) Show that, however large the initial speed u may be, the maximum height the ball can reach is always less than $\frac{\pi v_0^2}{4g}$.

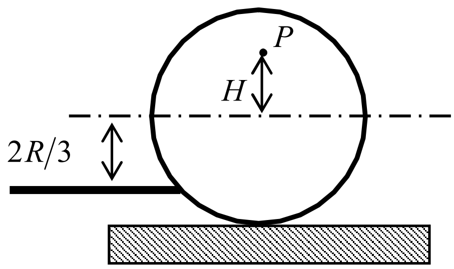
(Hint: $\int \frac{dx}{x^2 + a^2} = \frac{1}{a}\tan^{-1}\left(\frac{x}{a}\right) + C$)

3. (20pts) A billiard ball of radius R is given a sharp horizontal blow by a cue stick. The hitting point is $2R/3$ below the centerline of the ball. The ball is at rest on the table initially and it obtains a speed v_0 just after the hitting. The coefficient of kinetic friction between the ball and the table is μ_k.

(a) What is the angular speed ω_0 of the ball just after the hitting?

(b) A point P that is at height H above the center of the ball is at rest relative to the table just after the hitting. Find H.

(c) What is the speed of the ball once it begins to roll without slipping?



4. (20pts) It is observed that the minimum distance of a comet of mass m_c from the sun is exactly the half of the radius of the earth's orbit around the sun, and its speed at that point is twice the orbital speed of the earth. Ignore the effects of the earth and other planets on the comet, assume the mass of the comet and the mass of the earth is negligible compared to the mass of the sun M_s, and assume the earth's orbit is circular with radius R and in the same plane of the comet's orbit.

(a) Find the orbital speed of the earth v_E in terms of the mass of the sun M_s, the radius of its orbit R and the gravitational constant G.

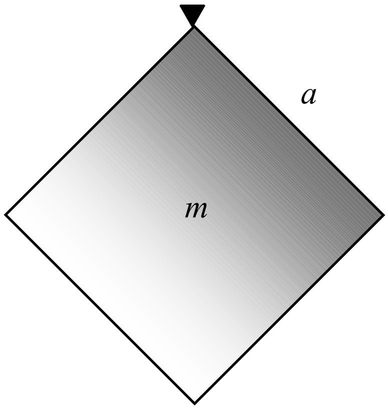
(b) What is the total energy of the comet? What type of orbit the comet travels in around the sun?

(c) Find the speed of the comet when it crosses the earth's orbit.

(d) Find the angle of the angle between the comet's orbit and the earth's orbit at which they cross.

5. (20 pts.) A stick of length L is initially standing vertically on the floor. If the stick falls over, with what angular velocity will it hit the floor? Assume that the end in contact with the floor does not slip.

6. (20pts) As shown in the figure, a thin square plate of mass m is suspended vertically at one of its corners in a way so that it can oscillate freely (left and right) in the plane of the square. The side length of the square is a .



(a) Write down the equation of rotational motion of the square plate.

(b) Find the period of small oscillations of the plate.

(c) If the square plate oscillates (back and forth) in the direction perpendicular to its plane, what is the period of small oscillations?